



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RL

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	5
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	04
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	08

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mrs. Sneha Yadav

Signature :

Date :

Q. 1) A robot is learning to move through a maze to reach a charging station. At each step, it receives a small or penalty depending on its movements. Instead of waiting until the end of the maze to learn whether its path was good or bad, the robot updates its learning after every move. Question - using the above situations, define temporal difference (TD) learning and identify how it can be applied in this problem. Also explore the parameters involved in the TD update rule in details.

→ Temporal difference (TD) learning is a reinforcement learning method where the agent.

- learns after every step
- updates values estimates using current reward + future value.
- combine ideas from monte carlo + dynamic programming.

- Applying to the maze problem -

- situation - Robot moves step-by-step in a maze
 - gets - small penalty
 - reward at charging station.

- TD update rule -

$$V(s) \leftarrow V(s) + \alpha [r + \gamma V(s') - V(s)]$$

- Explanation of parameters -

- $V(s)$ - current estimate of value of state s
- α (alpha) - controls how much we update
- r - reward received after taking action.
- γ - importance of future rewards.
- $V(s')$ - estimated value of next state
- TD error - $\delta = r + \gamma V(s') - V(s)$

Q. 2) A robot is moving inside a grid world to reach a target location. It completes several full paths and records the total reward obtained in each path. question - find how monte carlo prediction can be applied to estimate the value of each grid position? explain the steps involved.

→ Monte carlo (MC) prediction -

- learns value of states using complete episodes
- Uses actual total reward (returns)
- Applying to grid world
- situation - robot completes full paths
 - records total reward of each path
 - Eg. start $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ Goal

S1 - Initialize values - set $V(s) = 0$

S2 - generate episodes - robot runs full episodes -

- Eg. A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ Goal

S3 - Compute returns (G) - for each state -

$$G_t = r_{t+1} + \gamma V_{t+2} + \dots$$

- Eg. from state A $\rightarrow$ total return + 5

S4 - Update value function - for each visited state -

$V(s) = \text{average of returns observed from } s$

- first visit vs every visit - first visit MC - update only
- every visit MC - update all